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AI-Driven Automated Answer Script Evaluation System: A Literature Review

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Abstract:

Descriptive handwritten exam replies are still evaluated in a subjective and time-consuming manner. Assessment must be automated immediately due to the increasing size of classes and the necessity for prompt feedback. A survey of recent developments in automated grading employing artificial intelligence (AI), optical character recognition (OCR), and natural language processing (NLP) is presented in this work. We examine ten significant works from 2022–2025 , contrast their methodologies, and point out any deficiencies. The motivation for our suggested intelligent evaluation system, which combines feedback generation, scoring logic, semantic natural language processing, and OCR, is also described in the study.

Keywords: Answer sheet evaluation, AI in education, OCR, NLP, machine learning, automated grading.

1. INTRODUCTION

Exam answer sheet grading by hand is time-consuming, inconsistent, and prone to mistakes. Fairness problems are caused by subjective biases, weariness, and evaluators' differences. A viable substitute for automating the assessment process, cutting down on turnaround time, and increasing accuracy is the use of artificial intelligence (AI) and machine learning (ML). Handwritten responses are digitized using OCR, answer semantics are assessed using NLP, and insights and adaptive scoring are provided using machine learning in modern systems.

Manually grading exam answer sheets takes a lot of time, is inconsistent, and is prone to errors. Subjective biases, fatigue, and disparities among evaluators are the root causes of fairness issues. Artificial intelligence (AI) and machine learning (ML) are a good alternative for automating the assessment process, reducing turnaround time, and improving accuracy. In contemporary systems, OCR is used to digitize handwritten responses, NLP is used to evaluate the semantics of the responses, and machine learning is used to provide insights and adaptive scoring.

PROBLEM STATEMENT

There are various problems with traditional grading systems:

- Inconsistency in Assessment: Fairness issues are introduced by human marking variability.
- Postponed Response: Grading by hand takes a lot of time and postpones learning feedback.
- Inability to Scale: For large classes or nationwide exams, the current systems are ineffective.

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- Minimal Customization: Feedback is frequently general and unspecific to each person's performance.
- Cognitive Stress on Teachers: Manual grading leads to evaluator fatigue and burnout.

OBJECTIVES

The purpose of our system is to:

- Utilize OCR to digitize handwritten response sheets while applying preprocessing filters including line segmentation, thresholding, and denoising.
- To assess responses, use NLP techniques such as semantic similarity, keyword extraction, and sentence embeddings.
- Use weighted rubrics, configurable criteria, and model responses to automatically score using cosine similarity.
- Provide dashboards and reports on individual feedback together with real-time performance data.
- Give instructor interfaces the ability to export results, submit model solutions, and modify scoring algorithm.

2. RELATED WORK

We now provide a summary of important current research on AI-based automated grading.

A. *AI-Based Grading of Handwritten Responses [1]*

To compare student responses to model responses, Aureo et al. created a system that combines OCR with Sentence-BERT (SBERT) embeddings. Semantic similarity measures measured the importance of the answers, and the OCR module was tuned for handwritten text. Compared to manual evaluation, the method showed increased grading speed and uniformity. Its lack of adaptability for user engagement and rubric customization, however, limited its application in a variety of educational contexts. **Comparison:** This is furthered by our system, which gives teachers the ability to create rubrics, modify weights, and access visual statistics to better comprehend grading results.

B. *Evaluating GPT-4 for Math Exams [2]*

The ability of GPT-4 to grade handwritten math tests was evaluated by Caraeni et al. To parse equations and explanations, they employed sophisticated OCR engines in conjunction with GPT-4's language comprehension capabilities. OCR errors on intricate symbols and formula layouts were among the difficulties, which affected the interpretability of GPT-4. The study emphasized the necessity of domain-specific preprocessing while highlighting the possibility of massive language models. **Comparison:** Prior to NLP evaluation, we improve the recognition of technical and mathematical information by using domain-aware OCR tuning and improved preprocessing filters.

C. *AI-PAT: Automated Answer Evaluation [3]*

Combining text transformers for semantic grading and vision transformers for picture input, Lin and Banerjee presented an end-to-end transformer pipeline for automated answer evaluation. Their method answered descriptive and multiple-choice questions, but it was less accurate for answers that were tailored to a particular specialty subject. Without retraining, adaptability to other disciplines was restricted. **Comparison:** Broader domain coverage is made possible without retraining big models because to our framework's support for instructor input for model replies and rubric norms.

D. *OCR and NLP Pipeline for Education [4]*

Sinha and Verma presented a modular pipeline that combines NLP modules for response matching and keyword extraction with OCR modules for handwritten text extraction. Although their system was efficient at processing text, it lacked interactive features that would have allowed teachers to dynamically modify parameters or keep an eye on grading in real time. **Comparison:** In order to close this gap, we improve usability by introducing dynamic dashboards and adaptable feedback generation modules.

E. DistilBERT for Short Answer Grading [5]

In contrast to full BERT, Thomas and Lee used the lightweight DistilBERT model to effectively assess brief, objective responses, allowing for quicker inference with less accuracy loss. Their strategy was restricted to brief responses and did not work well for longer, more detailed answers or responses involving several participants. **Comparison:** Our method, which uses transformer models optimized for semantic similarity across disciplines, provides both brief and lengthy descriptive responses.

F. SmartCheck UPSC Evaluator [6]

Mehta's SmartCheck technology assessed the coherence and relevance of the content of UPSC test responses by combining big language models with OCR. Despite encouraging outcomes, accuracy was decreased by frequent OCR mismatches brought on by handwriting variability, requiring manual corrections. **Comparison:** To improve OCR accuracy and decrease error propagation, our system incorporates enhanced preprocessing for line recognition, segmentation, and noise reduction.

G. Automated Essay Scoring Using CNN [7]

Convolutional neural networks (CNNs) were used by Desai and Rao to assess essay coherence and structure and extract language elements. Although the model worked well for typed essays, it had trouble with handwriting variability and was not integrated with OCR for unprocessed handwritten data. **Comparison:** Our hybrid pipeline evaluates unprocessed handwritten descriptive responses by combining transformer-based semantic models with OCR preparation.

H. Multimodal Grading Systems [8]

Zhang et al. presented a multimodal grading system that combines textual semantic analysis, image texture analysis, and visual data from handwriting strokes. Improved grading accuracy and resilience to handwriting irregularities resulted from the combination of modalities. **Comparison:** Our future roadmap calls for multimodal integration to take advantage of complimentary handwriting and image data, even though we are now concentrating on text-based evaluation.

I. Adaptive Feedback Generation in Automated Grading [9]

In order to create individualized recommendations and educational materials, Kumar and Singh created an adaptive feedback architecture that clusters student responses and applies machine learning. Through focused interventions, this strategy increased student engagement and improved learning outcomes. **Comparison:** With intentions to integrate adaptive and tailored feedback generating in further rounds, our system offers basic feedback reports.

J. Transformer-Based Long Answer Scoring [10]

Gupta et al. scored long-form responses by combining the evaluation of coherence, content relevance, and grammatical correctness using transformer architectures that were refined on extensive datasets. Their approach showed resilience to different answer styles and produced state-of-the-art outcomes on multiple benchmark datasets. **Comparison:** Our method allows for modification for various evaluation settings by utilizing comparable transformer techniques with extra user options for rubric alterations.

3. RESEARCH GAP

Despite the potential of many ways, there are still significant challenges:

- Incomplete integration of the entire process, from scanned input to the creation of the final report.

- No manual overrides or customizable scoring policies are available.
- Real-time visual dashboards and feedback analytics are not supported.
- Restricted cross-disciplinary generalization (e.g., STEM vs. humanities)

4. CONCLUSION

Although AI-based grading systems have advanced, this literature analysis shows that comprehensive, approachable, and adaptable solutions are still lacking. To close these gaps and improve academic evaluation, our study suggests a strong architecture that combines OCR, NLP, semantic scoring, and feedback generation.

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